HK IMO Training 2021-2022
Problem Set 8
For 20 November 2021

Problem 29. Solve the equation $3^m + 271 = 125 \cdot 2^n$ in positive integers.

Problem 30. There is a 5×100 table. We choose n of the cells and call them *special cells*. If every cell has at most 2 adjacent special cells (cells sharing common sides), find the greatest possible value of n.

Problem 31. Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(xy-1) + f(x)f(y) = 2xy - 1$$

for all $x, y \in \mathbb{R}$.

Problem 32. Let $\triangle ABC$ be an acute triangle with circumcentre O and centroid G. Define the following.

- D is the midpoint of BC
- E is a point inside $\triangle ABC$ such that $AE \perp BC$ and $\angle BEC = 90°$
- F is the intersection of EG and OD
- P is a point on BC such that $FP /\!/ OB$
- Q is a point on BC such that $FQ /\!/ OC$
- X is a point on AB such that $XP \perp BC$
- Y is a point on AC such that $YQ \perp BC$
- Γ is the circle tangent to OB and OC at B and C respectively
- Ω is the circumcircle of $\triangle AXY$

Prove that Γ and Ω are tangent to each other.